

Introduction to Deep Learning and Neural Network

PTP Program



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Background

Biological Neuron (1)

ANNs were first introduced back in 1943 by the neurophysiologist Warren McCulloch and the mathematician Walter Pitts. In their paper, McCulloch and Pitts presented **a simplified computational model of how biological neurons might work together** in animal brains to perform complex computations using propositional logic.

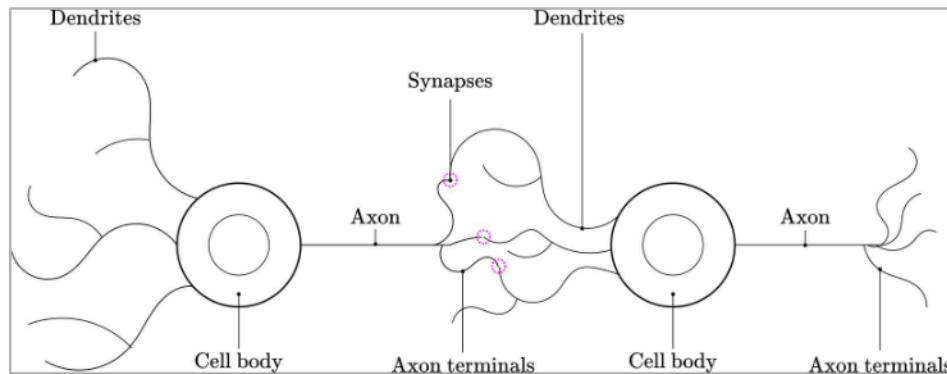
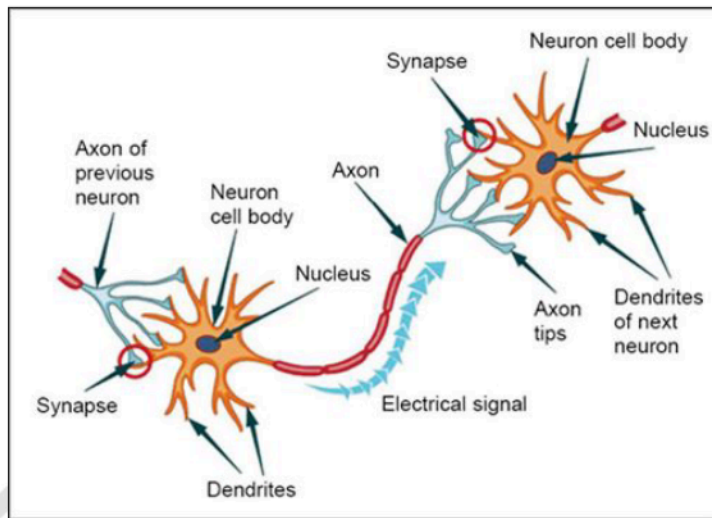
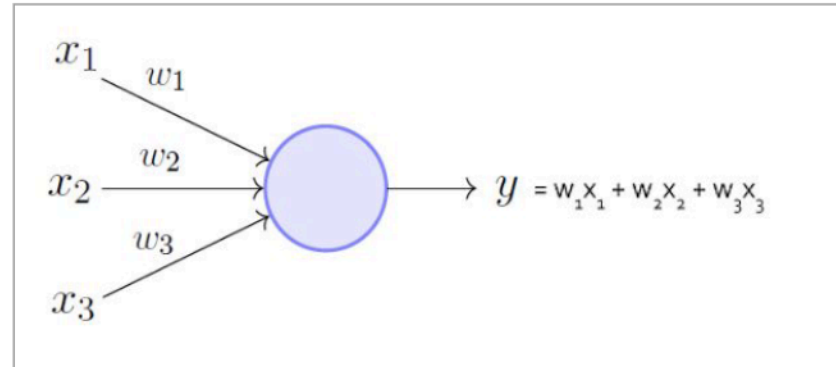
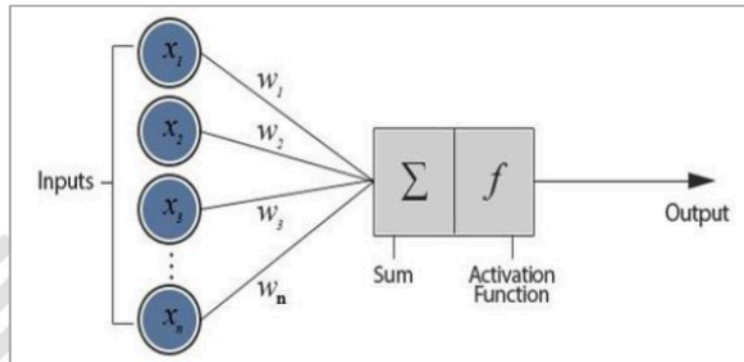
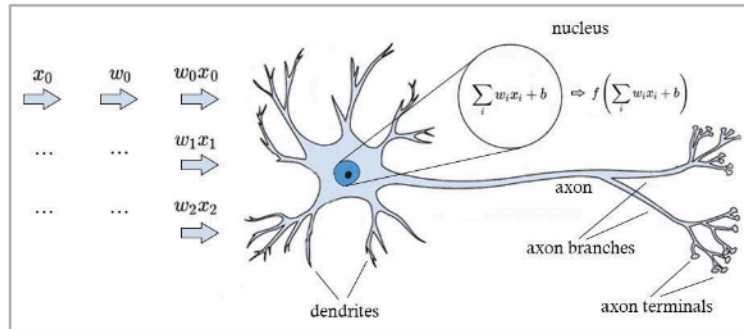


Illustration of biological neurons.

Background

Biological Neuron (2)

Proposed simplification of biological neurons.



Perceptron

Definition (1)

A perceptron is a type of artificial neural network architecture, often considered the simplest form of a neural network.

Here are the characteristics a perceptron:

1. Input Layer

The perceptron takes a set of input values, typically represented as a vector. Each input is associated with a weight.

2. Weights

Weights are assigned to each input and represent the strength/importance of the connection between the input and the perceptron.

3. Aggregation Function

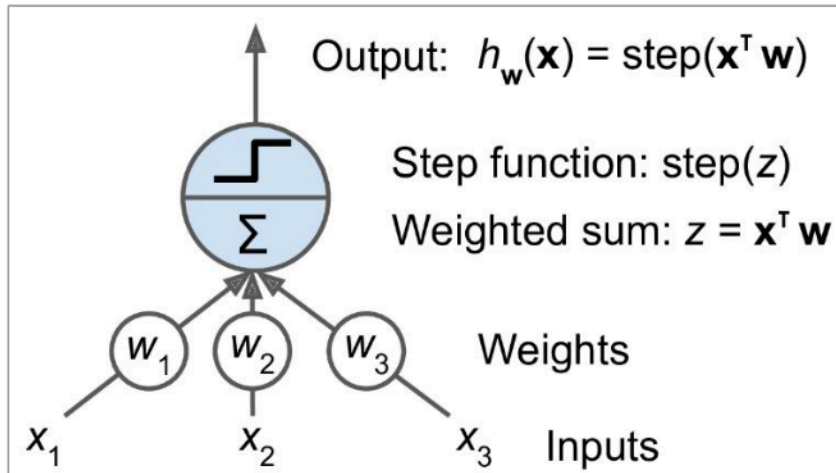
The perceptron computes a weighted sum of its inputs.

Perceptron

Definition (2)

4. Activation Function

The aggregation result is then passed through an activation function, which determines whether the perceptron should fire (produce an output of 1) or not (produce an output of 0). The most common activation function used in perceptrons is the step function.



Common **step functions** used in Perceptron (assuming threshold = 0)

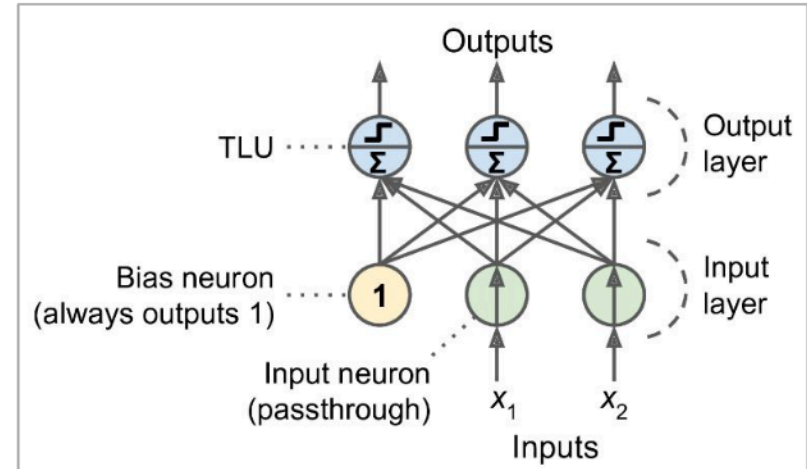
$$\text{Heaviside}(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$

$$\text{Sgn}(z) = \begin{cases} -1 & \text{if } z < 0 \\ 0 & \text{if } z = 0 \\ 1 & \text{if } z > 0 \end{cases}$$

Perceptron

Definition (3)

5. Output Layer
The perceptron produces a set of output layer.
6. Fully connected layer (dense layer)
When all the neurons in a layer are connected to every neuron in the previous layer.
7. Perceptron usually consist of 1 input layer and 1 output layer.



Architecture of a perceptron with 2 input neurons, 1 bias neuron, and 3 output neurons.

Multilayer Perceptron (MLP)

Definition

Problem with Perceptron is the network incapable of solving some trivial problems (e.g., the Exclusive Or (XOR) classification problem and more complex problem. This is because perceptron will make a linear separable hyperplane.

Multilayer Perceptron (MLP) was developed to tackle this limitation.

- It is a neural network that will create non-linear hyperplane.
- In MLP, there is layer called Hidden Layer.
- Hidden Layers introduce non-linearity into the network making them suitable for a wide range of tasks.
- MLP is composed of one input layer, one or more hidden layers, and one output layer.
- Every layer except the output layer includes a bias neuron and is fully connected to the next layer.
- The term Deep Neural Network (DNN) or Deep Learning is an ANN that contains a deep stack of hidden layers.

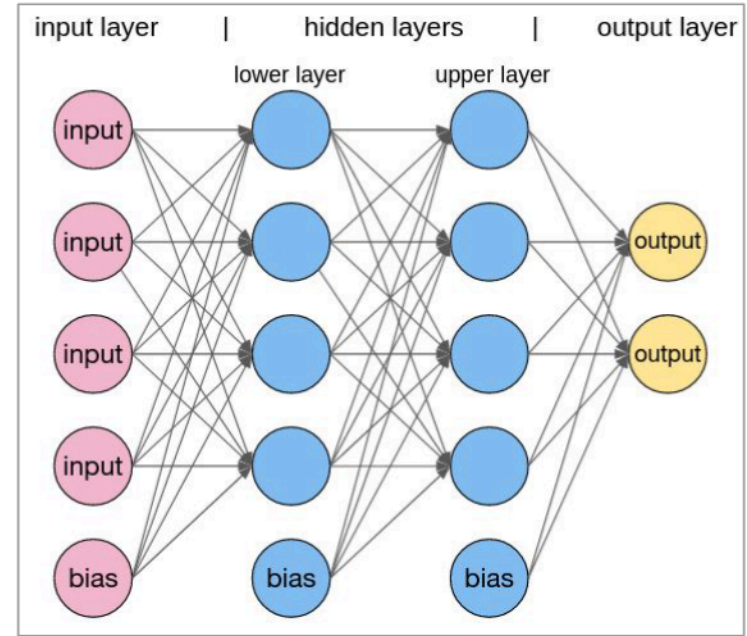
You can visit [TensorFlow Playground](#) to illustrate how MLP works on different types of data.

Multilayer Perceptron (MLP)

Example

The image beside is architecture of a MLP with configuration :

- 1 Output Layer (contains 4 input neurons)
- 2 Hidden Layers (each layer contains 4 neurons)
- 1 Output Layer (contains 2 output neurons)



Multilayer Perceptron (MLP)

Regression Tasks

- **Predict single value** (e.g., the price of a house), you just need **one output neuron**.
- **Predict multiple values at once, one output neuron per value**. For example, predict 2D coordinates, you need two output neurons.
- In general, for regression using MLP, you don't want to use any activation function for the output neurons. They are free to output any range of values.
- You can use ReLU function in the output neurons if the output value must be positive.

Multilayer Perceptron (MLP)

Classification Tasks

Binary Classification

- You just need a single output neuron.
- The output will be a probability between 0 and 1.
- You need a threshold value to convert the probability into a binary class.

Multilabel Classification

- Example : email classification (spam/not spam and urgent/non urgent)
- You would need two output neurons, with one output neuron for spam/not spam and one output neuron for urgent/non urgent.

Multiclass Classification

- One output neuron per class.
- Use softmax for activation function in output neurons.

Multilayer Perceptron (MLP)

Architectural Recommendations

Aspect		Regression	Binary Classification	Multiclass Classification
Input Layer	Number of neuron	One per input feature		
Hidden Layers	Number of layers	Depend on the problem (trial and error)		
	Number of neurons	Depend on the problem (trial and error)		
	Activation Function	ReLU; Leaky ReLU; SELU; ELU		
Output Layer	Number of neurons	1	1	1 per class
	Activation Function	Linear; Sigmoid; RELU (if positive outputs)	Sigmoid	Softmax
Loss Function		MSE; MAE	Binary Cross Entropy	Categorical Cross Entropy; Sparse Categorical Cross Entropy

Implementation

TensorFlow (1)

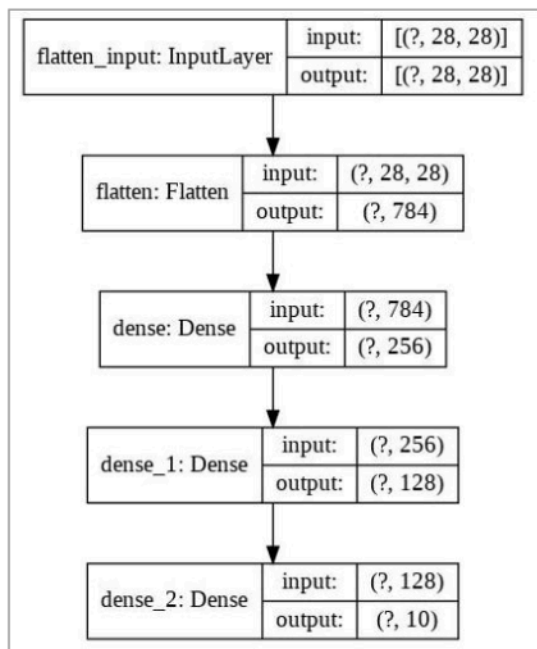
To building and designing Artificial Neural Network in Python, we can use library called [TensorFlow](#). In TensorFlow, there are two different approaches to create the architecture :

Sequential API	Functional API
Simple to use	Complex to implement
Only 1 input layer	Can have multiple input layer
Only 1 output layer	Can have multiple output layer
Layer-by-layer	Can sharing of layers
Output from a layer can only be input for one next layer	Output from a layer can be input for multi next layers
Rigid architecture	Flexible architecture

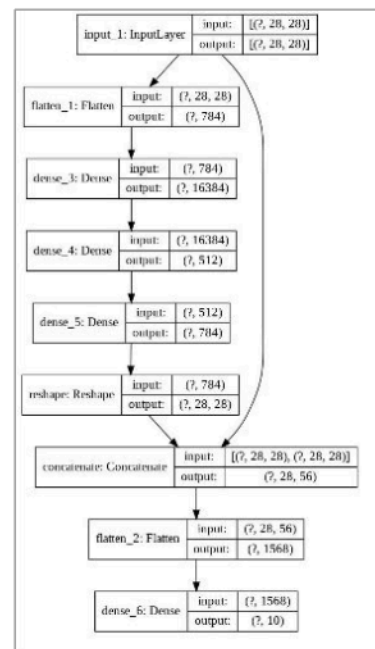
Implementation

TensorFlow (2)

Sequential API



Functional API



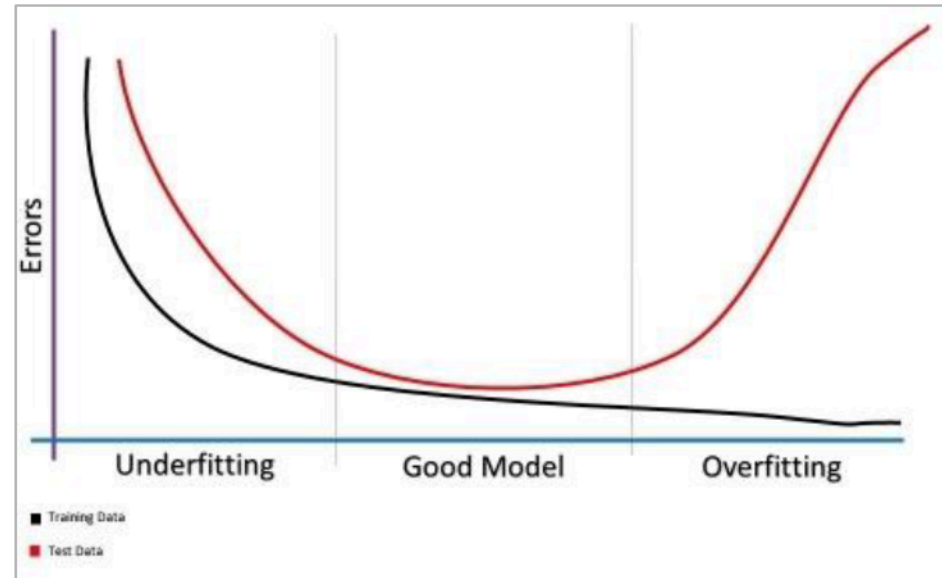
Learning Curve

Definition

Learning curve is a plot that shows the performance of the model incrementally over time. **It is common to create dual learning curves for ANN during training on both the training-set and validation-set.**

Description of plot beside :

- *x-axis* : epochs/iterations and *y-axis* : loss/error.
- *Black-line* : performance against train-set.
- *Red-line* : performance against validation-set.
- **The training process should be stopped when reach critical point (the validation error (red-line) trend changes from descending to ascending).**
- **Underfit** : if we stop the process **before** critical points.
- **Overfit** : if we stop the process **after** critical points.

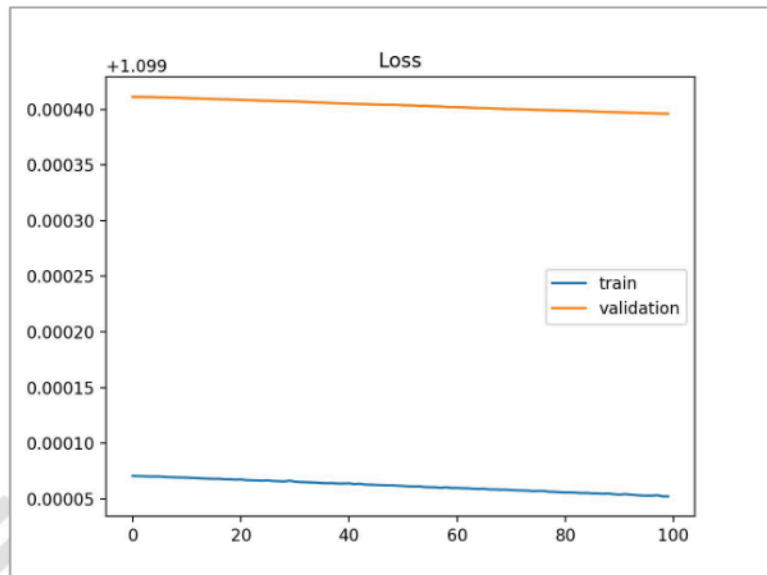


Learning Curve

Underfit

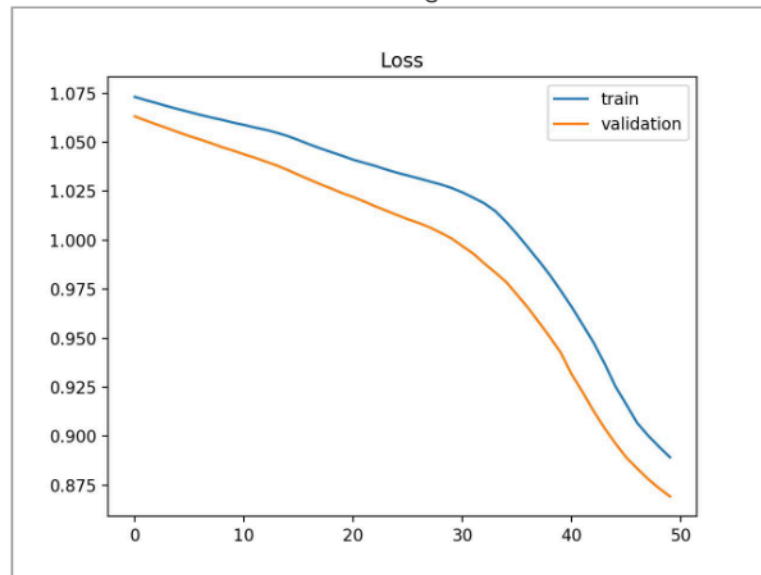
Case 1 :

the training loss remains flat regardless of training



Case 2 :

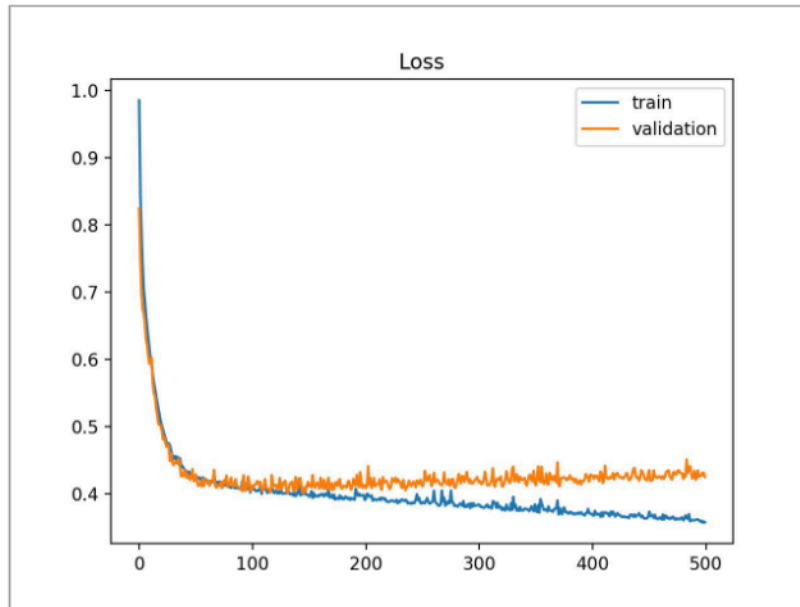
The training loss continues to decrease until the end of training



Learning Curve

Overfit

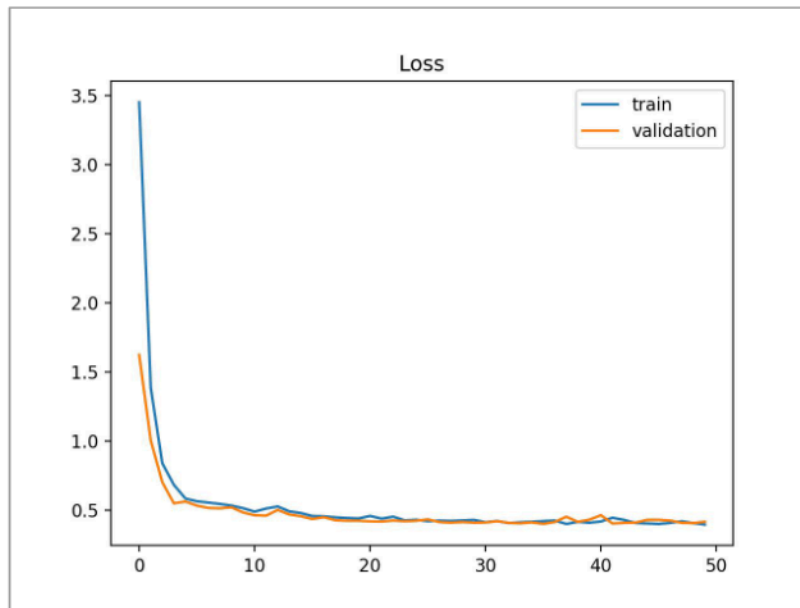
Overfit occurs when : Training loss continues to decrease as the epoch increases **AND** validation loss decreases to a point and begins increasing again



Learning Curve

Good-fit

Good-fit occurs when : Training loss continues to decrease to a point of stability **AND** validation loss decreases to a point of stability has a small gap with the training loss.



External References

Binary Classification



[Training](#)

[Inference](#)

Multiclass Classification



[Visit Here](#)



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Natural Language Processing

Definition

Natural Language Processing (NLP) is the sub-field of Artificial Intelligence (AI) that focuses on the interaction between computers and human language. NLP enables computers to understand, interpret, and generate human language (whether the language is spoken or written) in a way that is both valuable and meaningful.

Example of NLP applications both in form of text and speech :

Text	Speech
Text Classification	Speech Recognition
Text Summarization	Transcription Service
Sentiment Analysis	Voice Sentiment Analysis
Chatbots	Voice Assistants
Machine Translation	Voice Biometrics

Natural Language Processing

Challenges (1)

Understanding human language presents several challenges such as :

1. **Ambiguity**

Words and phrases can have multiple meanings depending on the context. For example, the word "bank" can refer to a financial institution or the side of a river.

2. **Coreference**

Coreference is the task of determining which words refer to the same entities in a text. For example, "I was eating bread while waiting for John at the train station. When he arrived, his mobile phone fell."
The words "John" and "he" are the same entity.

3. **Idioms and Figurative Language**

Idiomatic expressions, metaphors, and figurative language can pose challenges as their meanings are not always literal. For example : "break a leg".

Natural Language Processing

Challenges (2)

4. Context Dependency

The meaning of a word or phrase can change depending on the surrounding words and sentences. Understanding the context in which language is used is crucial for accurate comprehension.

5. Syntax and Grammar

Natural languages have complex grammatical rules and structures. This include informal words and multiple words that have the same meaning.

6. Sarcasm

Sarcasm is saying one thing but the meaning is the opposite.

Addressing these challenges is an active area of research in the fields of Artificial Intelligence and Natural Language Processing.

Terminology

Terminology

Here are terminologies related to NLP :

→ **Corpus**

A collection of text documents that used to train or evaluate the model.

→ **Document**

An individual piece of text within a corpus.

→ **Token**

The elements that make up a document. It can be words, hashtags, emojis, etc.

→ **Tokenization**

The process of breaking down text/document into individual tokens.

→ **Vocabulary**

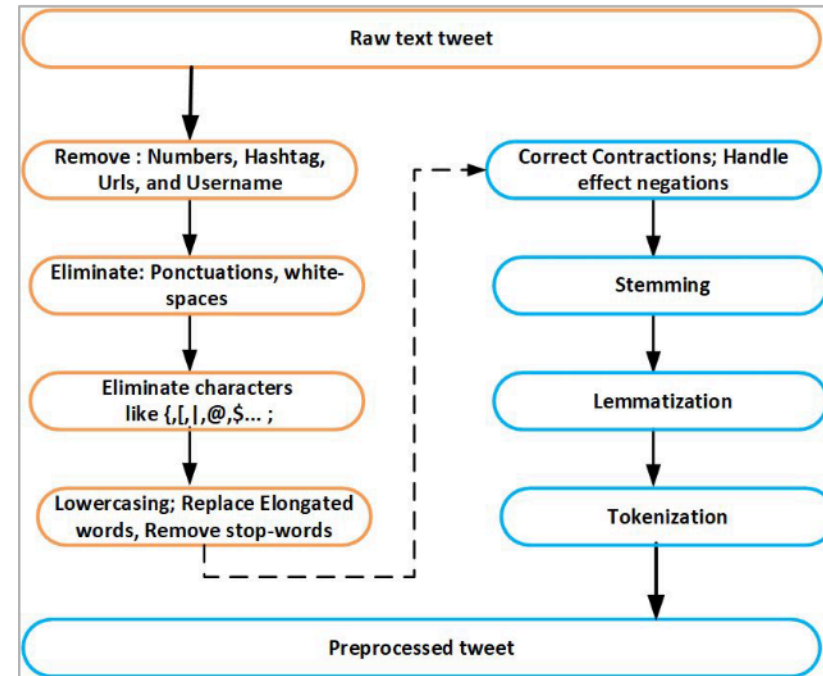
Unique tokens.

Text Preprocessing

Methods

There are numerous ways to do text preprocessing in NLP. It really **depends on the problem and the dataset**. Beside is an image to illustrate text preprocessing.

Our goal is to have as little vocab as possible and as representative as possible.



Sequential Data

Definition (1)

Let's take a look at dataset below. What happens to the dataset if we change the order of features between `Subject` and `Popularity` ?

	Year	Length	Title	Subject	Actor	Actress	Director	Popularity	Awards
0	1990	111.0	Tie Me Up! Tie Me Down!	Comedy	Bandaras Antonio	Abril Victoria	Almodovar Pedro	68.0	No
1	1991	113.0	High Heels	Comedy	Bos Miguel	Abril Victoria	Almodovar Pedro	68.0	No
2	1983	104.0	Dead Zone The	Horror	Walken Christopher	Adams Brooke	Cronenberg David	79.0	No
3	1979	122.0	Cuba	Action	Connery Sean	Adams Brooke	Lester Richard	6.0	No
4	1978	94.0	Days of Heaven	Drama	Gere Richard	Adams Brooke	Malick Terrence	14.0	No

	Year	Length	Title	Popularity	Actor	Actress	Director	Subject	Awards
0	1990	111.0	Tie Me Up! Tie Me Down!	68.0	Bandaras Antonio	Abril Victoria	Almodovar Pedro	Comedy	No
1	1991	113.0	High Heels	68.0	Bos Miguel	Abril Victoria	Almodovar Pedro	Comedy	No
2	1983	104.0	Dead Zone The	79.0	Walken Christopher	Adams Brooke	Cronenberg David	Horror	No
3	1979	122.0	Cuba	6.0	Connery Sean	Adams Brooke	Lester Richard	Action	No
4	1978	94.0	Days of Heaven	14.0	Gere Richard	Adams Brooke	Malick Terrence	Drama	No

Sequential Data

Definition (2)

Let's try another dataset :

Example :

Before	After
I read a book in the library	I read a book in library the
The food is good , not bad at all	The food is bad , not good at all
The capital city of Indonesia is Jakarta and the capital city of Japan is Tokyo	The capital city of Indonesia is Tokyo and the capital city of Japan is Jakarta

If we change the positions of the two words in each of the sentences above, the sentence becomes meaningless, changes the fact, and has the opposite meaning from its original meaning. This kind of data is called Sequential Data.

Sequential Data is any kind of data where the order of data points/features are matters. If there are two features or more are swapped, it will change the meaning of the data and even become meaningless. Example : text, image, movie, song, etc.

Sequential Data

Example (1)

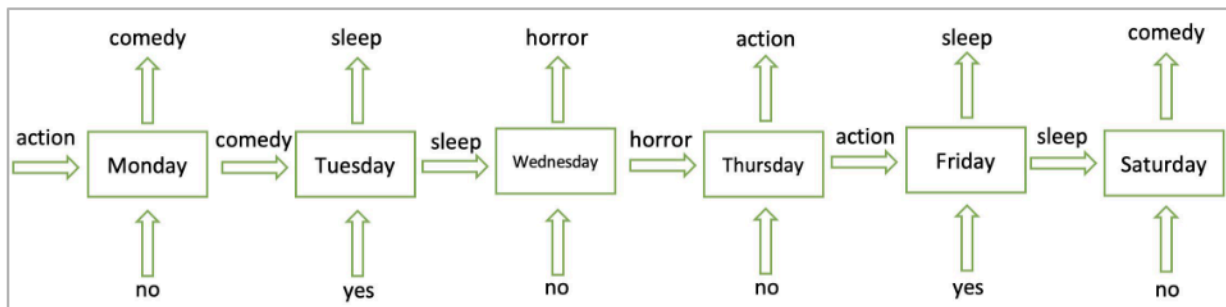
Let's say there is an employee named Bambang. Bambang really enjoys watching movies every night.

- Bambang likes movies with genres : action, comedy, and horror.
- Every night, he can only watch 1 movie.
- If he watches an action movie tonight, tomorrow he will watch a comedy movie, and the day after tomorrow he will watch a horror movie.
- He can only watch a movie after returning from work.
- If he doesn't work overtime, he can watch a movie after work. If he works overtime, he will go to sleep immediately after work.

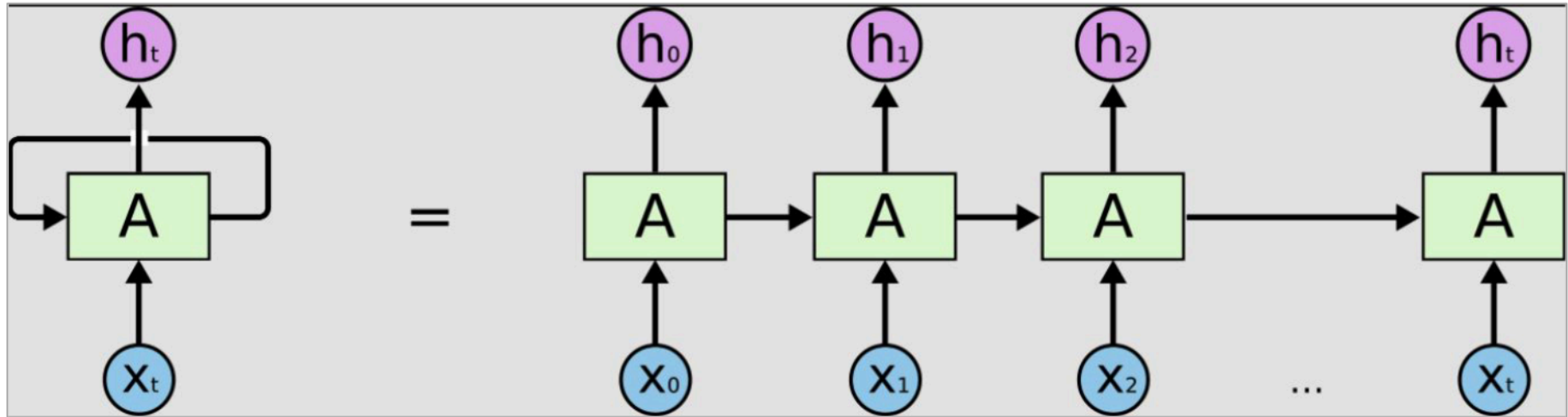
Sequential Data

Example (2)

	Day						
	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	action	comedy	sleep	horror	action	sleep	comedy
Overtime	no	no	yes	no	no	yes	no



Recurrent Neuron Illustration



A recurrent neuron (left) unrolled through time (right)

Recurrent Neuron

Gradient Problem Effect

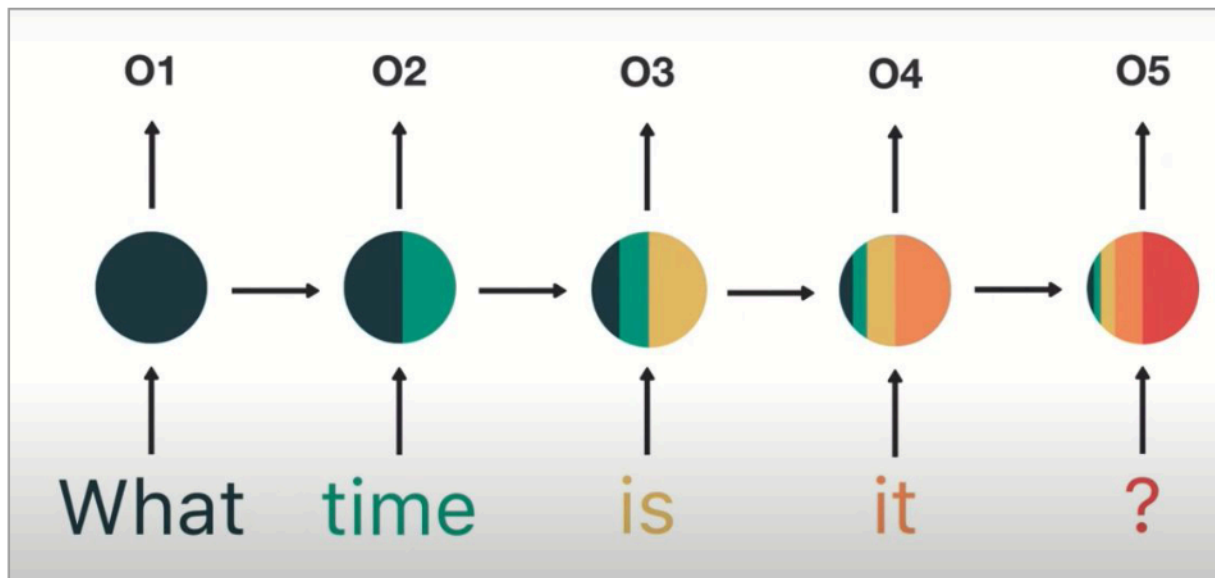
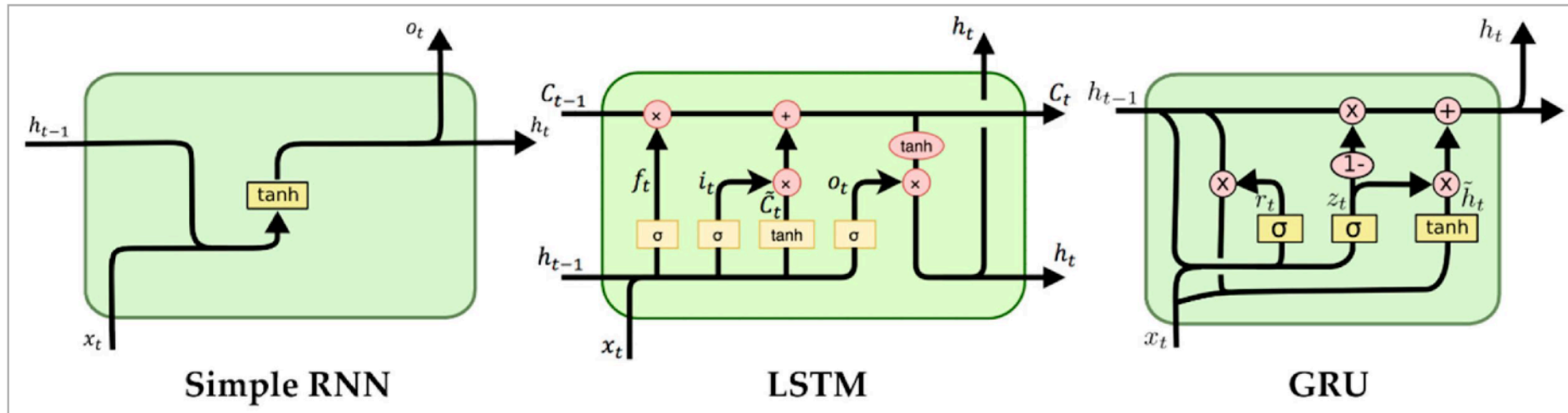


Illustration of gradient problem effect in Recurrent Neural Network

Recurrent Neuron

Types of Recurrent Neurons



Types of recurrent neurons.

Thank You



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